

## SUOMUSSALMI PESIO, Finland

WMO: 028890

Lat: 64.9314N

Lon: 28.7467E

Elev: 739

StdP: 14.31

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -23.1      | -17.2 | -28.0                         | 1.2 | -22.5 | -21.5 | 1.7 | -16.9 | 14.3                  | 21.8 | 13.3 | 20.2 | 2.1                   | 300  | 0.659 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.2                   | 77.6            | 63.4 | 73.8 | 61.2 | 70.4 | 59.8 | 66.2                | 73.6 | 63.8 | 70.5 | 61.6 | 67.9 | 5.6                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 63.7                            | 90.8 | 69.1 | 61.2 | 82.9 | 66.8 | 58.9 | 76.4 | 64.4 | 31.3          | 73.2 | 29.4 | 70.7 | 27.9 | 67.7 | 75.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 13.7          | 12.0 | 10.7 | -28.4                      | 81.2 | 6.2                | 4.3 | -32.9                                              | 84.3 | -36.5      | 86.8 | -40.0      | 89.2 | -44.6      | 92.3 |    |    |
| (5)               |      |      | -28.2                      | 68.6 | 6.1                | 3.6 | -32.6                                              | 71.2 | -36.2      | 73.3 | -39.6      | 75.3 | -44.1      | 77.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 34.5   | 11.3  | 12.7  | 20.2  | 32.0  | 44.2  | 52.5  | 59.5  | 55.3  | 46.4  | 34.5  | 25.4  | 18.1  |  |  |
|                                                                        | DBStd       | 18.88  | 14.53 | 13.40 | 10.56 | 7.65  | 8.25  | 6.63  | 6.09  | 5.90  | 6.05  | 7.40  | 9.47  | 12.83 |  |  |
|                                                                        | HDD50       | 6346   | 1201  | 1044  | 925   | 541   | 217   | 47    | 5     | 20    | 138   | 480   | 738   | 990   |  |  |
|                                                                        | HDD65       | 11175  | 1666  | 1464  | 1390  | 991   | 645   | 379   | 189   | 306   | 557   | 945   | 1188  | 1455  |  |  |
|                                                                        | CDD50       | 673    | 0     | 0     | 0     | 0     | 38    | 121   | 300   | 182   | 31    | 1     | 0     | 0     |  |  |
|                                                                        | CDD65       | 27     | 0     | 0     | 0     | 0     | 1     | 3     | 19    | 4     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 295    | 0     | 0     | 0     | 0     | 17    | 57    | 180   | 41    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 32     | 0     | 0     | 0     | 0     | 0     | 7     | 22    | 3     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 5.1    | 4.4   | 4.7   | 5.7   | 5.4   | 5.9   | 5.4   | 4.7   | 4.6   | 5.0   | 5.3   | 5.1   | 4.8   |  |  |
|                                                                        | PrecAvg     | 26.3   | 1.8   | 1.5   | 1.3   | 1.3   | 2.2   | 2.4   | 3.5   | 3.1   | 2.6   | 2.4   | 2.1   | 2.1   |  |  |
| (15) Precipitation                                                     | PrecMax     | 36.7   | 3.5   | 2.8   | 2.2   | 2.2   | 3.7   | 5.4   | 5.8   | 5.9   | 4.6   | 5.9   | 4.2   | 4.4   |  |  |
|                                                                        | PrecMin     | 19.3   | 0.3   | 0.3   | 0.3   | 0.4   | 0.7   | 0.6   | 0.9   | 1.0   | 1.1   | 0.4   | 0.8   | 0.3   |  |  |
|                                                                        | PrecStd     | 4.0    | 0.8   | 0.7   | 0.5   | 0.6   | 0.9   | 1.1   | 1.2   | 1.3   | 1.1   | 1.3   | 0.9   | 1.1   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 36.3  | 38.5  | 45.8  | 58.8  | 76.5  | 80.7  | 82.7  | 78.9  | 66.9  | 52.7  | 45.8  | 38.0  |  |  |
|                                                                        |             | MCWB   | 35.3  | 36.2  | 37.8  | 46.3  | 55.9  | 62.2  | 66.7  | 64.7  | 58.6  | 50.0  | 45.2  | 37.1  |  |  |
|                                                                        | 2%          | DB     | 33.0  | 34.6  | 39.5  | 51.8  | 71.3  | 74.2  | 79.0  | 73.7  | 62.3  | 49.2  | 42.1  | 35.7  |  |  |
|                                                                        |             | MCWB   | 32.7  | 33.4  | 34.6  | 42.4  | 55.9  | 60.4  | 65.3  | 62.9  | 55.6  | 47.7  | 41.3  | 34.9  |  |  |
|                                                                        | 5%          | DB     | 31.7  | 32.2  | 36.0  | 47.8  | 64.9  | 69.6  | 75.2  | 69.9  | 58.9  | 46.5  | 38.5  | 33.8  |  |  |
|                                                                        |             | MCWB   | 31.5  | 31.7  | 32.7  | 39.3  | 52.1  | 56.8  | 63.0  | 60.6  | 54.0  | 45.2  | 38.0  | 33.2  |  |  |
|                                                                        | 10%         | DB     | 29.7  | 30.1  | 33.7  | 44.2  | 59.5  | 65.9  | 71.7  | 66.6  | 56.1  | 44.4  | 35.8  | 32.3  |  |  |
|                                                                        |             | MCWB   | 29.3  | 29.4  | 31.7  | 37.7  | 48.5  | 55.3  | 61.4  | 59.3  | 52.7  | 43.4  | 35.4  | 32.0  |  |  |
|                                                                        | 0.4%        | WB     | 35.5  | 36.5  | 38.3  | 47.7  | 59.3  | 65.9  | 70.1  | 68.3  | 60.3  | 50.9  | 45.6  | 37.8  |  |  |
|                                                                        |             | MCDB   | 36.7  | 38.1  | 44.2  | 55.2  | 70.9  | 75.1  | 77.0  | 75.1  | 64.9  | 52.1  | 45.9  | 38.0  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 32.5  | 33.7  | 35.3  | 43.5  | 56.4  | 62.2  | 67.6  | 64.5  | 57.1  | 47.8  | 41.3  | 35.1  |  |  |
|                                                                        |             | MCDB   | 32.7  | 34.5  | 38.0  | 50.6  | 67.9  | 70.5  | 74.9  | 70.6  | 60.7  | 48.8  | 41.9  | 35.5  |  |  |
|                                                                        | 5%          | WB     | 31.6  | 31.8  | 33.6  | 40.4  | 53.2  | 59.3  | 65.2  | 62.2  | 54.9  | 45.4  | 38.0  | 33.3  |  |  |
|                                                                        |             | MCDB   | 31.7  | 32.2  | 35.4  | 46.1  | 63.8  | 66.9  | 72.3  | 67.8  | 57.8  | 46.2  | 38.5  | 33.7  |  |  |
|                                                                        | 10%         | WB     | 29.3  | 29.6  | 32.2  | 38.2  | 50.1  | 56.8  | 63.0  | 60.2  | 53.1  | 43.5  | 35.5  | 32.1  |  |  |
|                                                                        |             | MCDB   | 29.6  | 30.1  | 33.4  | 43.0  | 57.7  | 63.4  | 69.5  | 65.2  | 55.5  | 44.4  | 35.8  | 32.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.8  | 13.8  | 17.6  | 17.3  | 18.1  | 17.4  | 17.2  | 15.9  | 12.8  | 8.1   | 7.9   | 10.4  |  |  |
|                                                                        |             | MCDBR  | 9.2   | 9.6   | 17.0  | 23.9  | 28.6  | 25.2  | 22.9  | 22.2  | 17.6  | 10.1  | 7.4   | 7.8   |  |  |
|                                                                        |             | MCWBR  | 9.3   | 9.2   | 13.3  | 14.7  | 14.7  | 12.2  | 10.4  | 12.1  | 11.6  | 8.5   | 7.4   | 7.7   |  |  |
|                                                                        |             | MCDBR  | 9.0   | 9.4   | 12.9  | 20.8  | 25.4  | 20.5  | 20.3  | 19.4  | 15.4  | 9.3   | 7.3   | 7.1   |  |  |
|                                                                        | 5% WB       | MCWBR  | 9.1   | 9.0   | 10.6  | 13.4  | 13.8  | 11.0  | 10.5  | 11.4  | 10.9  | 8.3   | 7.3   | 7.0   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.198 | 0.268 | 0.298 | 0.332 | 0.339 | 0.339 | 0.361 | 0.340 | 0.317 | 0.293 | 0.350 | 0.198 |  |  |
|                                                                        | taud        |        | 1.931 | 2.114 | 2.163 | 2.257 | 2.464 | 2.486 | 2.444 | 2.481 | 2.532 | 2.386 | 2.658 | 1.958 |  |  |
|                                                                        | Ebn at Noon |        | 100   | 195   | 245   | 265   | 276   | 278   | 267   | 263   | 246   | 192   | 95    | 28    |  |  |
|                                                                        | Edh at Noon |        | 8     | 19    | 28    | 33    | 30    | 30    | 30    | 26    | 20    | 15    | 7     | 3     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 39    | 199   | 651   | 1190  | 1476  | 1628  | 1529  | 1131  | 618   | 218   | 49    | 12    |  |  |
|                                                                        | RadStd      |        | 4     | 41    | 59    | 121   | 154   | 115   | 156   | 112   | 65    | 33    | 10    | 1     |  |  |

## Historical Trends

|                              |  | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (25 neighbors) |  | +1.03   | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page